

PUMaC Problems I Wrote

July 3, 2026

Scope. These are the 48 unique PUMaC problem statements I wrote, identified from the acknowledgements in the accessible solution files. I merged duplicate Division A/B appearances in the headings. I omitted solutions and answer keys. I did not include 2024 because the archive page did not provide solution PDFs I could check.

Contents

1	2020	2
2	2021	2
3	2022	3
4	2023	8

1 2020

Problem 1. 2020 Algebra A #2 / Algebra B #4

Let C denote the curve

$$y^2 = \frac{x(x+1)(2x+1)}{6}.$$

The points $(1/2, a)$, (b, c) , and $(24, d)$ lie on C and are collinear, and $ad < 0$. Given that b, c are rational numbers, find $100b^2 + c^2$.

Problem 2. 2020 Geometry A #2 / Geometry B #4

Hexagon $ABCDEF$ has an inscribed circle Ω tangent to each of its sides. If $AB = 12$, $\angle FAB = 120^\circ$, $\angle ABC = 150^\circ$, and the radius of Ω can be written as $m + \sqrt{n}$ for positive integers m, n , find $m + n$.

Problem 3. 2020 Combinatorics A #8

Let $f(k)$ denote the number of triples (a, b, c) of positive integers satisfying $a + b + c = 2020$ with $k - 1$ not dividing a , k not dividing b , and $k + 1$ not dividing c . Find the product of all integers k in the range $3 \leq k \leq 20$ such that $k + 1$ divides $f(k)$.

2 2021

Problem 4. 2021 Algebra B #1

Let x, y be distinct positive real numbers satisfying

$$\frac{1}{\sqrt{x+y} - \sqrt{x-y}} + \frac{1}{\sqrt{x+y} + \sqrt{x-y}} = \frac{x}{\sqrt{y^3}}.$$

If

$$\frac{x}{y} = \frac{a + \sqrt{b}}{c}$$

for positive integers a, b, c with $\gcd(a, c) = 1$, find $a + b + c$.

Problem 5. 2021 Algebra A #8

Consider the sequence of Fibonacci numbers $F_0, F_1, F_2, \dots$, given by $F_0 = F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$. Define the sequence $x_0, x_1, x_2, \dots$ by $x_0 = 1$ and

$$x_{k+1} = x_k^2 + F_{2^k}^2$$

for $k \geq 0$. Define the sequence $y_0, y_1, y_2, \dots$ by $y_0 = 1$ and

$$y_{k+1} = 2x_k y_k - y_k^2$$

for $k \geq 0$. If

$$\sum_{k=0}^{\infty} \frac{1}{y_k} = \frac{a - \sqrt{b}}{c}$$

for positive integers a, b, c with $\gcd(a, c) = 1$, find $a + b + c$.

Problem 6. 2021 Combinatorics A #5 / Combinatorics B #7

A Princeton slot machine has 100 pictures, each equally likely to occur. One is a picture of a tiger. Alice and Bob independently use the slot machine, and each repeatedly makes independent plays. Alice keeps playing until she sees a tiger, at which point she stops. Similarly, Bob keeps playing until he sees a tiger. Given that Bob plays twice as much as Alice, let the expected number of plays for Alice be a/b , with a, b relatively prime positive integers. Find the remainder when $a + b$ is divided by 1000.

Problem 7. 2021 Combinatorics A #6 / Combinatorics B #8

Alice, Bob, and Carol are playing a game. Each turn, one of them says one of the three players' names, chosen from $\{\text{Alice}, \text{Bob}, \text{Carol}\}$ uniformly at random. Alice goes first, Bob goes second, Carol goes third, and they repeat in that order. Let E be the expected number of names that have been said when, for the first time, all three names have been said twice. If $E = m/n$ for relatively prime positive integers m, n , find $m + n$. Include the last name to be said twice in your count.

Problem 8. 2021 Team #3

Let

$$f(N) = N \left(\frac{9}{10} \right)^N,$$

and let m/n denote the maximum value of $f(N)$ as N ranges over the positive integers. If m, n are relatively prime positive integers, find the remainder when $m + n$ is divided by 1000.

Problem 9. 2021 Team #5

Given a real number t with $0 < t < 1$, define the real-valued function

$$f(t, \theta) = \sum_{n=-\infty}^{\infty} t^{|n|} \omega^n, \quad \omega = e^{i\theta} = \cos \theta + i \sin \theta.$$

For $\theta \in [0, 2\pi)$, the polar curve $r(\theta) = f(t, \theta)$ traces out an ellipse E_t with a horizontal major axis whose left focus is at the origin. Let $A(t)$ be the area of the ellipse E_t . Let

$$A\left(\frac{1}{2}\right) = \frac{a\pi}{b},$$

where a, b are relatively prime positive integers. Find $100a + b$.

3 2022

Problem 10. 2022 Algebra A #2 / Algebra B #4

Let $P(x, y)$ be a polynomial with real coefficients in the variables x, y that is not identically zero. Suppose that

$$P(\lfloor 2a \rfloor, \lfloor 3a \rfloor) = 0$$

for all real numbers a . If P has the minimum possible degree and the coefficient of the monomial y is 4, find the coefficient of x^2y^2 in P .

The degree of a monomial $x^m y^n$ is $m + n$. The degree of a polynomial $P(x, y)$ is the maximum degree of any of its monomials.

Problem 11. 2022 Algebra A #6 / Algebra B #8

Let x, y, z be positive real numbers satisfying

$$4x^2 - 2xy + y^2 = 64, \quad y^2 - 3yz + 3z^2 = 36, \quad 4x^2 + 3z^2 = 49.$$

If the maximum possible value of $2xy + yz - 4zx$ can be expressed as $\sqrt{n}$ for some positive integer n , find n .

Problem 12. 2022 Algebra A #7

For a positive integer $n \geq 1$, let

$$a_n = \left\lfloor \sqrt[3]{n} + \frac{1}{2} \right\rfloor.$$

Given a positive integer $N \geq 1$, let F_N denote the set of positive integers $n \geq 1$ such that $a_n \leq N$. Let

$$S_N = \sum_{n \in F_N} \frac{1}{a_n^2}.$$

As $N \rightarrow \infty$, the quantity $S_N - 3N$ tends to $a\pi^2/b$ for relatively prime positive integers a, b . Given that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6},$$

find $a + b$.

Problem 13. 2022 Algebra A #8

The function f sends sequences to sequences in the following way: given a sequence $\{a_n\}_{n=0}^{\infty}$ of real numbers, f sends $\{a_n\}_{n=0}^{\infty}$ to the sequence $\{b_n\}_{n=0}^{\infty}$, where

$$b_n = \sum_{k=0}^n a_k \binom{n}{k}$$

for all $n \geq 0$. Let $\{F_n\}_{n=0}^{\infty}$ be the Fibonacci sequence, defined by $F_0 = 0$, $F_1 = 1$, and $F_{n+2} = F_{n+1} + F_n$ for all $n \geq 0$. Let $\{c_n\}_{n=0}^{\infty}$ denote the sequence obtained by applying the function f to the sequence $\{F_n\}_{n=0}^{\infty}$ 2022 times. Find $c_5 \pmod{1000}$.

Problem 14. 2022 Combinatorics A #1 / Combinatorics B #3

In the country of PUMaC-land, there are 5 villages and 3 cities. Vedant is building roads between the 8 settlements according to the following rules:

1. There is at most one road between any two settlements.
2. Any city has exactly three roads connected to it.

3. Any village has exactly one road connected to it.
4. Any two settlements are connected by a path of roads.

In how many ways can Vedant build the roads?

Problem 15. 2022 Combinatorics A #7

Kelvin has a set of eight vertices. For each pair of distinct vertices, Kelvin independently draws an edge between them with probability $p \in (0, 1)$. A set S of four distinct vertices is called good if there exists an edge between v and w for all $v, w \in S$ with $v \neq w$. The variance of the number of good sets can be expressed as a polynomial $f(p)$ in the variable p . Find the sum of the absolute values of the coefficients of $f(p)$.

The variance of a random variable X is defined as $\mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Problem 16. 2022 Geometry A #1 / Geometry B #3

Let Γ be the circle centered at $(0, 0)$ with radius $2022\sqrt{3}$. Circle Ω is centered on the x -axis, passes through the point $A = (6066, 0)$, and intersects Γ orthogonally at the point $P = (x, y)$ with $y > 0$. If the length of the minor arc AP on Ω can be expressed as $m\pi/n$ for relatively prime positive integers m, n , find $m + n$.

Two circles are said to intersect orthogonally at a point P if the tangent lines at P form a right angle.

Problem 17. 2022 Geometry A #5 / Geometry B #7

Let $\triangle ABC$ be a triangle with $AB = 5$, $BC = 8$, and $CA = 7$. Let the center of the A -excircle be O , and let the A -excircle touch lines BC , CA , and AB at points X, Y, Z , respectively. Let h_1, h_2, h_3 denote the distances from O to lines XY , YZ , and ZX , respectively. If

$$h_1^2 + h_2^2 + h_3^2$$

can be written as m/n for relatively prime positive integers m, n , find $m + n$.

Problem 18. 2022 Geometry A #7

Let $\triangle ABC$ be a triangle with $BC = 7$, $CA = 6$, and $AB = 5$. Let I be the incenter of $\triangle ABC$. Let the incircle of $\triangle ABC$ touch sides BC , CA , and AB at points D, E, F . Let the circumcircle of $\triangle AEF$ meet the circumcircle of $\triangle ABC$ for a second time at point $X \neq A$. Let P denote the intersection of XI and EF . If the product $XP \cdot IP$ can be written as m/n for relatively prime positive integers m, n , find $m + n$.

Problem 19. 2022 Geometry A #8

Let $\triangle ABC$ have sidelengths $BC = 7$, $CA = 8$, and $AB = 9$, and let Ω denote the circumcircle of $\triangle ABC$. Let circles $\omega_A, \omega_B, \omega_C$ be internally tangent to the minor arcs BC , CA , and AB of Ω , respectively, and tangent to the segments BC , CA , and AB at points X, Y, Z , respectively. Suppose that

$$\frac{BX}{XC} = \frac{CY}{YA} = \frac{AZ}{ZB} = \frac{1}{2}.$$

Let t_{AB} be the length of the common external tangent of ω_A and ω_B , let t_{BC} be the length of the common external tangent of ω_B and ω_C , and let t_{CA} be the length of the common external tangent of ω_C and ω_A . If $t_{AB} + t_{BC} + t_{CA}$ can be expressed as m/n for relatively prime positive integers m, n , find $m + n$.

Problem 20. 2022 Number Theory A #1 / Number Theory B #3

Find the sum of all prime numbers p such that p divides

$$(p^2 + p + 20)^{p^2+p+2} + 4(p^2 + p + 22)^{p^2-p+4}.$$

Problem 21. 2022 Number Theory A #3 / Number Theory B #5

Given $k \geq 1$, let p_k denote the k th smallest prime number. If N is the number of ordered 4-tuples (a, b, c, d) of positive integers satisfying

$$abcd = \prod_{k=1}^{2023} p_k$$

with $a < b$ and $c < d$, find $N \pmod{1000}$.

Problem 22. 2022 Number Theory A #5 / Number Theory B #7

A positive integer $\ell \geq 2$ is called sweet if there exists a positive integer $n \geq 10$ such that when the leftmost nonzero decimal digit of n is deleted, the resulting number m satisfies $n = m^\ell$. Let S denote the set of all sweet numbers ℓ . If

$$\sum_{\ell \in S} \frac{1}{\ell - 1} = \frac{A}{B}$$

for relatively prime positive integers A, B , find $A + B$.

Problem 23. 2022 Number Theory A #6 / Number Theory B #8

Given a positive integer ℓ , define the sequence $\{a_n^{(\ell)}\}_{n=1}^\infty$ by

$$a_n^{(\ell)} = \left\lfloor n + \sqrt[\ell]{n} + \frac{1}{2} \right\rfloor$$

for all positive integers n . Let S denote the set of positive integers that appear in all three of the sequences $\{a_n^{(2)}\}_{n=1}^\infty$, $\{a_n^{(3)}\}_{n=1}^\infty$, and $\{a_n^{(4)}\}_{n=1}^\infty$. Find the sum of the elements of S that lie in the interval $[1, 100]$.

Problem 24. 2022 Team #1

Let $b, c \in \mathbb{R}$ satisfy $b \in (0, 1)$ and $c > 0$. Let A, B denote the points of intersection of the line $y = bx + c$ with $y = |x|$, and let O denote the origin of $\mathbb{R}^2$. Let $f(b, c)$ denote the area of triangle $\triangle OAB$. Let $k_0 = 1/2022$, and for $n \geq 1$ let $k_n = k_{n-1}^2$. If

$$\sum_{n=1}^{\infty} f(k_n, k_{n-1})$$

can be written as p/q for relatively prime positive integers p, q , find the remainder when $p + q$ is divided by 1000.

Problem 25. 2022 Team #7

Pick x, y, z to be real numbers satisfying

$$(-x + y + z)^2 - \frac{1}{3} = 4(y - z)^2,$$

$$(x - y + z)^2 - \frac{1}{4} = 4(z - x)^2,$$

and

$$(x + y - z)^2 - \frac{1}{5} = 4(x - y)^2.$$

If the value of $xy + yz + zx$ can be written as p/q for relatively prime positive integers p, q , find $p + q$.

Problem 26. 2022 Team #8

Ryan Alweiss storms into the Fine Hall common room with a gigantic eraser and erases all integers n in the interval $[2, 728]$ such that 3^t does not divide $n!$, where

$$t = \left\lceil \frac{n-3}{2} \right\rceil.$$

Find the sum of the leftover integers in that interval modulo 1000.

Problem 27. 2022 Team #9

In the complex plane, let z_1, z_2, z_3 be the roots of the polynomial

$$p(x) = x^3 - ax^2 + bx - ab.$$

Find the number of integers n between 1 and 500 inclusive that are expressible as

$$z_1^4 + z_2^4 + z_3^4$$

for some choice of positive integers a, b .

Problem 28. 2022 Team #13

Of all functions $h : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{\geq 0}$, choose one satisfying

$$h(ab) = ah(b) + bh(a)$$

for all $a, b \in \mathbb{Z}_{>0}$ and $h(p) = p$ for all prime numbers p . Find the sum of all positive integers $n \leq 100$ such that $h(n) = 4n$.

Problem 29. 2022 Team #14

Let $\triangle ABC$ be a triangle. Let Q be a point in the interior of $\triangle ABC$, and let X, Y, Z denote the feet of the altitudes from Q to sides BC, CA , and AB , respectively. Suppose that $BC = 15$,

$\angle ABC = 60^\circ$, $BZ = 8$, $ZQ = 6$, and $\angle QCA = 30^\circ$. Let line QX intersect the circumcircle of $\triangle XYZ$ at the point $W \neq X$. If the ratio WY/WZ can be expressed as p/q for relatively prime positive integers p, q , find $p + q$.

Problem 30. 2022 Team #15

Subsets S of the first 35 positive integers $\{1, 2, 3, \dots, 35\}$ are called contrived if S has size 4 and the sum of the squares of the elements of S is divisible by 7. Find the number of contrived sets.

4 2023

Problem 31. 2023 Algebra B #2

The sum

$$\sum_{m=1}^{2023} \frac{2m}{m^4 + m^2 + 1}$$

can be expressed as a/b for relatively prime positive integers a, b . Find the remainder when $a + b$ is divided by 1000.

Problem 32. 2023 Algebra A #1 / Algebra B #3

Let a, b, c, d, e, f be real numbers such that

$$a^2 + b^2 + c^2 = 14, \quad d^2 + e^2 + f^2 = 77, \quad ad + be + cf = 32.$$

Find

$$(bf - ce)^2 + (cd - af)^2 + (ae - bd)^2.$$

Problem 33. 2023 Algebra A #3 / Algebra B #5

Let $P(x)$ be a polynomial with integer coefficients satisfying

$$(x^2 + 1)P(x - 1) = (x^2 - 10x + 26)P(x)$$

for all real numbers x . Find the sum of all possible values of $P(0)$ between 1 and 5000, inclusive.

Problem 34. 2023 Algebra A #5 / Algebra B #7

Compute

$$\sum_{k=0}^{10} \left(3 + 2 \cos \left(\frac{2\pi k}{11} \right) \right)^{10} \pmod{100}.$$

Problem 35. 2023 Algebra A #6 / Algebra B #8

A polynomial

$$p(x) = \sum_{j=1}^{2n-1} a_j x^j$$

with real coefficients is called mountainous if $n \geq 2$ and there exists a real number k such that the polynomial's coefficients satisfy $a_1 = 1$,

$$a_{j+1} - a_j = k \quad \text{for } 1 \leq j \leq n-1,$$

and

$$a_{j+1} - a_j = -k \quad \text{for } n \leq j \leq 2n-2.$$

We call k the step size of $p(x)$. A real number k is called good if there exists a mountainous polynomial $p(x)$ with step size k such that $p(-3) = 0$. Let S be the sum of all good numbers k satisfying $k \geq 5$ or $k \leq 3$. If $S = b/c$ for relatively prime positive integers b, c , find $b + c$.

Problem 36. 2023 Combinatorics B #2

Amir enters Fine Hall and sees the number 2 written on a blackboard. Amir can perform the following operation: he flips a coin, and if it is heads, he replaces the number x on the blackboard with $3x + 1$; otherwise, he replaces x with $\lfloor x/3 \rfloor$. If Amir performs this operation four times, let m/n denote the expected number of times that he writes the digit 1 on the blackboard, where m, n are relatively prime positive integers. Find $m + n$.

Problem 37. 2023 Geometry A #2 / Geometry B #4

Let $\triangle ABC$ be an isosceles triangle with $AB = AC = \sqrt{7}$ and $BC = 1$. Let G be the centroid of $\triangle ABC$. Given $j \in \{0, 1, 2\}$, let T_j denote the triangle obtained by rotating $\triangle ABC$ about G by $2\pi j/3$ radians. Let P denote the intersection of the interiors of triangles T_0, T_1, T_2 . If K denotes the area of P , then $K^2 = a/b$ for relatively prime positive integers a, b . Find $a + b$.

Problem 38. 2023 Geometry A #7

Let $\triangle ABC$ be a triangle with $\angle BAC = 90^\circ$, $\angle ABC = 60^\circ$, $\angle BCA = 30^\circ$, and $BC = 4$. Let the incircle of $\triangle ABC$ meet sides BC , CA , and AB at points A_0, B_0, C_0 , respectively, and let I be the incenter of $\triangle ABC$. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles $\triangle B_0IC_0$, $\triangle C_0IA_0$, and $\triangle A_0IB_0$, respectively.

Construct triangle T_A as follows: let A_0B_0 meet ω_B for the second time at $A_1 \neq A_0$, let A_0C_0 meet ω_C for the second time at $A_2 \neq A_0$, and let T_A denote the triangle $\triangle A_0A_1A_2$. Construct triangles T_B, T_C similarly. If the sum of the areas of triangles T_A, T_B, T_C equals $\sqrt{m} - n$ for positive integers m, n , find $m + n$.

Problem 39. 2023 Geometry A #8

Let $\triangle ABC$ be a triangle with $AB = 4$ and $AC = 7/2$. Let ω denote the A -excircle of $\triangle ABC$. Let ω touch lines AB and AC at the points D and E , respectively. Let Ω denote the circumcircle of $\triangle ADE$. Consider the line ℓ parallel to BC such that ℓ is tangent to ω at a point F and such that ℓ does not intersect Ω . Let ℓ intersect lines AB and AC at the points X and Y , respectively, with $XY = 18$ and $AX = 16$. Let the perpendicular bisector of XY meet the circumcircle of $\triangle AXY$ at P, Q , where the distance from P to F is smaller than the distance from Q to F . Let ray $\overrightarrow{PF}$ meet Ω for the first time at the point Z . If $PZ^2 = m/n$ for relatively prime positive integers m, n , find $m + n$.

Problem 40. 2023 Number Theory A #1 / Number Theory B #3

Find the integer x for which

$$135^3 + 138^3 = x^3 - 1.$$

Problem 41. 2023 Number Theory A #2 / Number Theory B #4

A number is called good if it can be written as the sum of the squares of three consecutive positive integers. A number is called excellent if it can be written as the sum of the squares of four consecutive positive integers. For instance, $14 = 1^2 + 2^2 + 3^2$ is good and $30 = 1^2 + 2^2 + 3^2 + 4^2$ is excellent. A good number G is called splendid if there exists an excellent number E such that

$$3G - E = 2025.$$

If the sum of all splendid numbers is S , find the remainder when S is divided by 1000.

Problem 42. 2023 Team #3

A quadratic polynomial $f(x)$ is called sparse if its degree is exactly 2, if it has integer coefficients, and if there exists a nonzero polynomial $g(x)$ with integer coefficients such that $f(x)g(x)$ has degree at most 3 and $f(x)g(x)$ has at most two nonzero coefficients. Find the number of sparse quadratics whose coefficients lie between 0 and 10, inclusive.

Problem 43. 2023 Team #5

Let S denote the set of all positive integers whose prime factors are elements of $\{2, 3, 5, 7, 11\}$. We include 1 in the set S . If

$$\sum_{q \in S} \frac{\varphi(q)}{q^2}$$

can be written as a/b for relatively prime positive integers a, b , find $a + b$. Here φ denotes Euler's totient function.

Problem 44. 2023 Team #6

Let $f(p)$ denote the number of ordered tuples $(x_1, x_2, \dots, x_p)$ of nonnegative integers satisfying

$$\sum_{i=1}^p x_i = 2022,$$

where $x_i \equiv i \pmod{p}$ for all $1 \leq i \leq p$. Find the remainder when

$$\sum_{p \in S} f(p)$$

is divided by 1000, where S denotes the set of all primes less than 2022.

Problem 45. 2023 Team #7

Alice, Bob, and Carol each independently roll a fair six-sided die and obtain the numbers a, b, c , respectively. They then compute the polynomial

$$f(x) = x^3 + px^2 + qx + r$$

with roots a, b, c . If the expected value of the sum of the squares of the coefficients of $f(x)$ is m/n for relatively prime positive integers m, n , find the remainder when $m + n$ is divided by 1000.

Problem 46. 2023 Team #8

Let $\triangle ABC$ be a triangle with sidelengths $AB = 5$, $BC = 7$, and $CA = 6$. Let D, E, F be the feet of the altitudes from A, B, C , respectively. Let L, M, N be the midpoints of sides BC , CA , and AB , respectively. If the area of the convex hexagon with vertices at D, E, F, L, M, N can be written as

$$\frac{x\sqrt{y}}{z}$$

for positive integers x, y, z with $\gcd(x, z) = 1$ and y square-free, find $x + y + z$.

Problem 47. 2023 Team #11

Let

$$f(z) = \frac{az + b}{cz + d}$$

for $a, b, c, d \in \mathbb{C}$. Suppose that $f(1) = i$, $f(2) = i^2$, and $f(3) = i^3$. If the real part of $f(4)$ can be written as m/n for relatively prime positive integers m, n , find $m^2 + n^2$.

Problem 48. 2023 Team #13

Let $\triangle TBD$ be a triangle with $TB = 6$, $BD = 8$, and $DT = 7$. Let I be the incenter of $\triangle TBD$, and let TI intersect the circumcircle of $\triangle TBD$ at $M \neq T$. Let lines TB and MD intersect at Y , and let lines TD and MB intersect at X . Let the circumcircles of $\triangle YBM$ and $\triangle XDM$ intersect at $Z \neq M$. If the area of $\triangle YBZ$ is x and the area of $\triangle XDZ$ is y , then the ratio x/y can be expressed as p/q , where p, q are relatively prime positive integers. Find $p + q$.